

A wildlife biologist surveyed the manatee population in Blue Springs State Park in 2009. He observed 326 manatees during that visit. He returned to the park each year since and found that, on average, there were 11 fewer manatees.

1. What are the quantities that are changing in this scenario?

Manatees per year

2. Define the variables m and y .

Answers will vary.

m represents the number of years since 2009.

y represents the number of manatees in Blue Springs State Park.

3. What is the parameter?

-11

4. What is the constant?

326

5. Which equation represents this scenario?

☐ $m = 326y - 11$

☐ $m = -11y + 326$

☒ $y = 326 - 11m$

☐ $y = -11 + 326y$

6. How many manatees were observed in 2015?

$m = 6, \quad y = 326 - 11(6)$

$= 326 - 66$

$= 260 \text{ manatees}$

7. Use the equation to find out how many years it would take for the manatee population to become extinct in Blue Springs if the trend observed by the wildlife biologist continues.

$0 = 326 - 11m$

$-326 = -11m$

$29\frac{7}{11} = m$

It will take approximately 30 years.

In 2021, the Florida Environmental Protection Agency (FL EPA) implemented a new program to feed lettuce to the manatees during the winter to try and stabilize the population. They predict that beginning that year, they can reverse the shrinking trend in the manatee population in Blue Springs so that it will now grow at the rate of 2 manatees per year.

8. Use the equation from the previous question to determine the manatee population in 2021.

$$\begin{aligned} 2021 - 2009 &= 12 = m & y &= 326 - 11(12) \\ & & &= 326 - 132 \\ & & &= 194 \text{ manatees} \end{aligned}$$

9. Write an equation using the variables m and y that represents the new predicted population of manatees after 2021.

$$y = 194 + 2m, m \text{ represents years after 2021}$$

10. Use your equation to predict in what year the manatee population will reach the same size it was in 2009.

$$\begin{aligned} 326 &= 194 + 2m \\ 132 &= 2m \\ 66 &= m \end{aligned}$$

The manatee population will reach 326, 66 years after 2021, in the year 2087.